

TIKANLIK, China

WMO: 517650

Lat: 40.6475N

Lon: 87.6943E

Elev: 847

StdP: 91.56

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-16.2	-23.7	0.5	-9.8	-21.7	0.6	-9.0	5.4	-3.0	4.7	-3.3	0.6	250	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	38.0	19.2	36.6	18.8	35.3	18.4	22.0	33.1	20.7	32.3	19.8	31.7	1.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	14.7	27.6	17.0	13.5	26.0	15.7	12.4	25.6	69.1	32.9	64.3	32.4	60.6	31.7	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.3	6.0	5.0														
(5)			DB	-19.6	40.8	1.5	1.2	-20.7	41.6	-21.6	42.3	-22.5	42.9	-23.6	43.8	
			WB	-20.0	23.9	1.5	1.9	-21.1	25.3	-22.0	26.4	-22.8	27.5	-23.9	28.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	-7.7	-1.1	7.8	15.9	21.6	26.2	27.8	26.5	20.7	12.1	3.0	-5.7
	DBStd	12.80	2.98	3.94	4.50	4.31	3.65	2.90	2.53	2.57	3.12	3.78	3.83	3.45
	HDD10.0	1682	550	311	97	7	0	0	0	0	0	22	209	487
	HDD18.3	3199	808	544	326	95	14	1	0	0	13	195	458	745
	CDD10.0	2527	0	0	29	184	358	484	552	510	321	87	1	0
	CDD18.3	1002	0	0	0	23	114	235	294	252	83	2	0	0
	CDH23.3	12921	0	0	8	384	1518	2988	3802	3140	1001	79	0	0
	CDH26.7	6743	0	0	0	108	676	1636	2194	1737	382	11	0	0
(14) Wind	WSAvg	1.8	1.1	1.4	2.1	2.6	2.6	2.2	2.0	1.8	1.8	1.5	1.3	1.2
(15) Precipitation	PrecAvg	36	0	0	1	2	3	7	11	6	3	1	1	0
	PrecMax	76	6	0	15	14	46	37	40	36	17	10	9	3
	PrecMin	12	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	21	1	0	3	4	8	9	11	8	5	3	3	1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	13.2	24.4	32.4	36.0	38.5	40.3	39.1	34.8	28.2	18.3	6.8
		MCWB	-1.1	4.1	9.8	15.0	16.5	18.5	19.8	19.6	17.5	13.5	7.4	1.0
	2%	DB	2.2	10.8	22.0	29.6	34.1	37.0	38.6	37.3	32.4	25.5	15.8	4.3
		MCWB	-2.4	3.0	8.7	13.0	15.6	18.2	19.7	19.0	16.5	12.2	6.4	-0.6
	5%	DB	0.4	9.0	19.8	27.4	32.3	35.6	37.0	35.8	30.7	23.3	13.5	2.3
		MCWB	-3.6	1.9	7.7	12.0	15.0	17.9	19.4	18.7	15.9	11.1	5.4	-1.8
	10%	DB	-1.3	6.9	17.4	25.5	30.6	34.2	35.5	34.3	28.9	21.0	10.8	0.3
		MCWB	-4.7	0.6	6.5	11.1	14.3	17.5	19.1	18.5	15.4	10.3	4.0	-3.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	4.5	10.5	15.2	18.9	22.5	24.0	23.6	20.6	14.2	7.9	1.4
		MCDB	4.4	12.4	23.0	31.1	31.8	32.9	34.4	32.6	29.3	25.0	17.3	6.1
	2%	WB	-2.3	3.2	8.9	13.6	16.7	20.3	22.3	21.7	18.4	13.1	6.5	-0.4
		MCDB	2.0	10.4	21.3	28.3	31.4	32.3	33.7	32.5	28.9	24.1	15.1	3.9
	5%	WB	-3.5	2.1	7.8	12.4	15.6	19.1	21.0	20.5	17.1	11.9	5.5	-1.7
		MCDB	0.2	8.7	19.3	26.3	30.5	31.8	32.8	31.8	27.9	22.0	13.2	2.0
	10%	WB	-4.7	0.7	6.6	11.4	14.7	18.2	20.0	19.5	16.1	10.7	4.1	-2.9
		MCDB	-1.5	6.7	17.1	24.4	28.9	31.3	32.1	31.2	27.0	20.0	10.6	0.2
(35) Mean Daily Temperature Range	MDBR		13.8	15.1	15.4	15.1	14.9	13.9	13.7	15.0	16.4	17.5	14.9	12.2
	5% DB	MCDBR	15.2	16.4	17.7	17.9	17.9	16.4	16.6	17.5	18.5	19.5	18.0	13.5
		MCWBR	11.2	10.1	9.0	7.7	6.9	6.3	6.3	6.9	7.5	8.5	10.0	9.3
	5% WB	MCDBR	14.9	16.1	17.5	16.6	16.0	14.4	14.2	15.6	17.0	18.7	17.7	13.1
		MCWBR	11.1	10.0	9.0	7.3	6.7	6.5	6.7	7.7	7.7	8.2	10.0	9.0
(40) Clear-Sky Solar Irradiance	taub		0.353	0.416	0.498	0.566	0.587	0.613	0.587	0.579	0.588	0.488	0.400	0.358
	taud		2.040	1.900	1.717	1.612	1.620	1.621	1.697	1.698	1.634	1.768	1.949	2.017
	Ebn at Noon		762	760	737	720	719	701	714	703	652	670	691	713
	Edh at Noon		117	157	212	252	255	256	235	228	228	177	128	111
(44) All-Sky Solar Radiation	RadAvg		2.59	3.60	4.81	5.86	6.83	7.05	6.81	6.33	5.48	4.19	2.94	2.21
	RadStd		0.18	0.13	0.18	0.31	0.30	0.40	0.24	0.30	0.18	0.13	0.12	0.21

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.40	N/A	N/A	+0.40	-1.09	-1.70	N/A	-110	+105	+58	
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	-0.64	N/A	N/A	N/A	+79	N/A	

Nomenclature: See separate page